# Dual-Stream CNN–Transformer Architecture for Remote Photoplethysmography-Based Heart Rate Estimation

## 1. Executive Summary

### Project Objective

The objective of this project is to develop a robust, end-to-end Remote Photoplethysmography (rPPG) system capable of estimating human heart rate from facial videos without any physical contact sensors. The system combines advanced computer vision, physiological signal processing, deep learning, and spectral analysis techniques to reconstruct blood volume pulse (BVP) signals directly from facial video sequences and estimate heart rate in beats per minute (BPM).

### Problem Statement

Traditional heart-rate monitoring systems require wearable sensors such as ECG electrodes, pulse oximeters, chest straps, or smartwatches. While these systems provide reliable measurements, they introduce discomfort, require direct skin contact, and may not be suitable for continuous, unobtrusive monitoring.

Remote Photoplethysmography (rPPG) addresses this limitation by estimating cardiovascular signals directly from RGB video. However, extracting physiological information from facial videos is highly challenging due to:

- Subject motion
- Facial expression changes
- Illumination variations
- Skin-tone diversity
- Camera noise
- Landmark instability
- Extremely low signal-to-noise ratio

The objective of this project is to design a learning-based framework capable of overcoming these challenges and producing reliable heart-rate estimates under realistic conditions.

## Main Contribution

The primary contribution of this work is the development of a Dual-Stream CNN–Transformer architecture that jointly models spatial, temporal, and spectral information present in facial video sequences.

The proposed framework introduces:

- Multi-scale CNN feature extraction
- CBAM attention mechanisms
- Dual motion–appearance streams
- Transformer-based temporal modeling
- Spectral attention enhancement
- Temporal convolution refinement
- Robust ROI extraction pipeline
- Physiological signal-aware BPM estimation

## Expected Outcomes

The system is expected to:

- Reconstruct high-quality rPPG waveforms
- Estimate heart rate accurately
- Improve robustness against motion artifacts
- Enhance physiological frequency preservation
- Generalize across diverse facial appearances
- Provide a foundation for future physiological monitoring systems

# 2. Introduction

## Photoplethysmography (PPG)

Photoplethysmography (PPG) is a non-invasive optical technique used to measure blood volume changes within biological tissues. During each cardiac cycle, the amount of blood flowing through peripheral vessels changes, producing subtle variations in light absorption and reflection.

These variations can be measured and transformed into physiological signals representing:

- Heart rate
- Heart rate variability
- Respiratory activity
- Cardiovascular health indicators

Conventional PPG systems typically use dedicated optical sensors consisting of LEDs and photodetectors attached to the skin.

## Remote Photoplethysmography (rPPG)

Remote Photoplethysmography extends traditional PPG by extracting blood volume pulse signals from video recordings without requiring physical contact.

Instead of specialized sensors, rPPG utilizes standard RGB cameras to detect subtle skin color fluctuations caused by pulsatile blood flow.

The principle relies on the observation that blood absorption characteristics vary throughout the cardiac cycle, producing imperceptible but measurable changes in facial skin color.

## Challenges of Camera-Based Heart Rate Estimation

### Motion Artifacts

Facial movements introduce intensity changes significantly larger than physiological pulse variations.

### Illumination Variations

Lighting changes can alter pixel intensities and mask subtle blood-flow signals.

### Skin Tone Diversity

Different melanin concentrations affect light reflection and physiological signal visibility.

### Landmark Instability

Face-tracking systems exhibit frame-to-frame landmark jitter, introducing artificial temporal fluctuations.

### Signal-to-Noise Ratio

The physiological signal occupies only a small fraction of observed pixel variations, making extraction extremely challenging.

## Why Deep Learning?

Traditional rPPG algorithms rely on handcrafted signal processing methods such as:

- POS
- CHROM
- ICA
- PCA

Although effective under controlled conditions, these approaches struggle under realistic environments.

Deep learning enables:

- Automatic feature learning
- Motion artifact suppression
- Illumination invariance
- Temporal pattern modeling
- End-to-end optimization

The proposed architecture leverages both CNNs and Transformers to capture complementary spatial and temporal physiological information.

# 3. System Overview

The proposed system follows a complete video-to-heart-rate pipeline.

## High-Level Architecture

```
Input Video
      │
      ▼
MediaPipe FaceMesh
      │
      ▼
Landmark Stabilization (EMA)
      │
      ▼
ROI Extraction
      │
      ▼
Skin Segmentation
      │
      ▼
CLAHE Enhancement
      │
      ▼
RGB Signal Acquisition
      │
      ▼
Dual Stream Generation
      │
      ├───────────────┐
      ▼               ▼
Motion Stream   Appearance Stream
      │               │
      ▼               ▼
CNN Encoder     CNN Encoder
      │               │
      └───────┬───────┘
              ▼
       Feature Fusion
              ▼
     Positional Encoding
              ▼
     Transformer Encoder
              ▼
      Spectral Attention
              ▼
 Temporal Convolution Block
              ▼
      rPPG Waveform Output
              ▼
      Signal Processing
              ▼
       BPM Estimation
```

# Pipeline Description

## Face Detection

MediaPipe FaceMesh detects facial landmarks in each frame.

## Landmark Stabilization

An Exponential Moving Average stabilizes landmark trajectories and reduces tracking jitter.

## ROI Extraction

Three physiological regions are extracted:

- Forehead
- Left Cheek
- Right Cheek

These regions are selected because they provide strong blood perfusion signals.

## Skin Segmentation

YCrCb-based skin masking removes background and non-skin pixels.

## CLAHE Enhancement

Contrast Limited Adaptive Histogram Equalization normalizes illumination conditions.

## Motion Stream

Temporal difference frames capture pulse-related changes and motion dynamics.

## Appearance Stream

Raw facial frames preserve texture, color, and anatomical information.

## CNN Feature Extraction

Each stream independently processes frames using a MultiScaleSpatialEncoder consisting of:

- Convolution blocks
- Depthwise separable convolutions
- CBAM attention modules
- Attention pooling

## Feature Fusion

Motion and appearance features are concatenated and projected into a unified latent representation.

## Transformer Encoder

A Transformer encoder models long-range temporal dependencies across the entire sequence.

## Spectral Attention

Frequency-domain attention selectively enhances physiologically meaningful components.

## Temporal Refinement

A residual temporal convolution block refines local waveform continuity.

## Signal Prediction

The network outputs a continuous rPPG waveform.

## BPM Estimation

The predicted signal undergoes:

- Detrending
- Bandpass filtering
- FFT analysis
- Harmonic rejection
- Parabolic interpolation

to produce the final heart-rate estimate.

# 4. Verified  Deep Learning Architecture

## Architecture Classification

Based on code inspection, the proposed model is formally classified as:

### Dual-Stream CNN–Transformer Architecture for Remote Photoplethysmography

The architecture combines:

1. Spatial CNN Feature Learning
2. Dual-Stream Physiological Representation
3. Transformer Temporal Modeling
4. Spectral Attention Enhancement
5. Temporal Convolution Refinement

## Motion Stream

Input:

```
Frame(t) − Frame(t−1)
```

Purpose:

- Capture subtle pulse-induced changes
- Suppress static background information
- Emphasize temporal physiological dynamics

## Appearance Stream

Input:

```
Raw Facial Frames
```

Purpose:

- Skin texture representation
- Facial morphology encoding
- Illumination context preservation

## Stream Fusion

The outputs of both streams are concatenated:

```
Motion Features
+
Appearance Features
```

and projected into a unified latent representation.

This design enables simultaneous learning of dynamic physiological changes and static facial characteristics.

## Why Dual Streams?

A single stream cannot effectively separate:

- Pulse-induced motion
- Appearance information

The dual-stream design allows each branch to specialize in a distinct information modality, improving robustness and physiological signal reconstruction.

# 5. CNN Architecture Analysis

The CNN component is implemented through the MultiScaleSpatialEncoder.

## CNN Structure

```
Input Frame
      │
      ▼
Stem Convolution
      │
      ▼
Multi-Scale Branches
      │
      ├── Scale 1
      ├── Scale 2
      └── Scale 3
      │
      ▼
CBAM Attention
      │
      ▼
Attention Pooling
      │
      ▼
Feature Concatenation
      │
      ▼
Projection Layer
```

The CNN learns:

- Blood-flow-related color fluctuations
- Skin texture patterns
- Illumination-invariant representations
- Multi-scale facial structures
- Pulse-sensitive facial regions

This spatial encoding stage provides rich physiological features that are later processed temporally by the Transformer.

# 6. ROI Extraction and Signal Acquisition

## Facial Landmark Detection

The proposed system employs MediaPipe FaceMesh for dense facial landmark localization. FaceMesh provides 468 facial landmarks per frame, enabling accurate facial geometry estimation under varying poses and expressions.

The primary objective of facial landmark detection is to establish anatomically meaningful regions that consistently contain strong physiological information.

Advantages include:

- Real-time performance
- High landmark density
- Robust facial tracking
- Cross-subject consistency

---

## Landmark Stabilization Using EMA

Raw landmark trajectories often exhibit frame-to-frame jitter due to tracking uncertainty.

To mitigate this issue, the system applies Exponential Moving Average (EMA) smoothing:

$$
L_t = \alpha X_t + (1-\alpha)L_{t-1}
$$

where:

- ($X_t$) = current landmark position
- ($L_t$) = stabilized landmark
- ($\alpha$) = smoothing factor

Benefits:

- Reduced landmark noise
- Stable ROI boundaries
- Improved temporal consistency
- Lower motion-induced artifacts

---

# Convex Hull ROI Generation

Rather than relying on rectangular bounding boxes, the system constructs convex hulls around selected facial landmarks.

Advantages:

- Better anatomical alignment
- Reduced background contamination
- More accurate skin coverage

---

# Physiological ROIs

## Forehead ROI

Characteristics:

- High skin visibility
- Minimal facial deformation
- Strong blood perfusion

## Left Cheek ROI

Characteristics:

- Rich vascular network
- High pulse signal strength

## Right Cheek ROI

Characteristics:

- Redundant physiological source
- Increased robustness against partial occlusion

Combining multiple ROIs improves signal stability and reduces dependence on a single facial region.

---

# Skin Segmentation Using YCrCb

Not all pixels within an ROI contain physiological information.

The system performs skin masking using the YCrCb color space:

- Y = luminance

- Cr = chrominance red
- Cb = chrominance blue

Skin pixels occupy a compact region in the Cr-Cb plane.

Benefits:

- Background suppression
- Removal of hair pixels
- Improved physiological signal purity

---

# CLAHE Enhancement

Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied to normalize illumination.

Benefits:

- Enhanced local contrast
- Reduced lighting sensitivity
- Improved signal visibility

---

# ROI Quality Assessment

The system evaluates ROI quality using:

## Skin Coverage

$$
Coverage = \frac{SkinPixels}{TotalPixels}
$$

## Sharpness Score

Estimated using Laplacian variance.

Higher values indicate:

- Better focus
- More reliable physiological information

---

# RGB Signal Extraction

For each ROI:

$$
[R_t, G_t, B_t]
$$

are computed as mean channel intensities.

These temporal RGB traces form the basis for traditional physiological signal extraction methods.

# 7. Signal Processing Pipeline

## Detrending

### Objective

Remove slow illumination drift and baseline wander.

Mathematically:

[
$x_d(t)=x(t)-\text{Trend}(t)$
]

Benefits:

- Stabilizes signal baseline
- Enhances periodic cardiac components

---

## Butterworth Bandpass Filtering

Human heart rate typically lies within:

[
40-200 BPM
]

Equivalent frequency range:

[
0.67 - 3.33 Hz
]

Bandpass filtering removes:

- Low-frequency drift
- High-frequency noise

Benefits:

- Improved physiological specificity
- Better SNR

---

# POS Method

Plane Orthogonal-to-Skin (POS) projects RGB signals into a physiological subspace.

Key idea:

Physiological signals occupy a different direction than illumination variations.

Benefits:

- Robust illumination suppression
- Improved pulse extraction

---

# CHROM Method

CHROM constructs chrominance combinations:

$$
X = 3R-2G
$$

$$
Y = 1.5R+G-1.5B
$$

Benefits:

- Motion robustness
- Enhanced pulse contrast

---

# FFT-Based BPM Estimation

Fast Fourier Transform converts the temporal signal into frequency space.

$$
X(f)=FFT(x(t))
$$

Dominant frequency corresponds to heart rate.

$$
BPM=60\times f_{peak}
$$

---

# Parabolic Peak Interpolation

FFT bins are discrete.

To improve precision:

$$
f_{refined}=f_{peak}+\Delta
$$

where $\Delta$ is estimated from neighboring bins.

Benefits:

- Sub-bin frequency estimation
- Reduced BPM quantization error

---

# Signal-to-Noise Ratio (SNR)

$$
SNR=10\log_{10}\left(\frac{P_{signal}}{P_{noise}}\right)
$$

Higher SNR indicates:

- Cleaner physiological waveform
- More reliable heart-rate estimates

# 8. Transformer Architecture Analysis

## Why CNNs Alone Are Insufficient

CNNs excel at:

- Spatial feature extraction
- Local pattern learning

However:

Heart-rate estimation requires modeling relationships spanning hundreds of frames.

CNN receptive fields remain limited.

---

## Positional Encoding

Transformers lack temporal awareness.

Sinusoidal positional encoding injects sequence order information:

$$PE(pos,2i)=\sin(pos/10000^{2i/d})$$

$$PE(pos,2i+1)=\cos(pos/10000^{2i/d})$$

Benefits:

- Sequence ordering
- Temporal structure preservation

---

## Multi-Head Self-Attention

For each frame:

$$Q=XW_Q$$

[
K=XW_K
]

[
V=XW_V
]

Attention:

**[**
**Attention(Q,K,V)**

Softmax\left(
\frac{QK^T}{\sqrt{d}}
\right)V
]

Benefits:

- Long-range dependency modeling
- Global temporal context

---

# Transformer Encoder

The implementation contains:

- Multi-head attention
- Feed-forward network
- Residual connections
- Layer normalization

Configuration:

- D Model = 128
- Heads = 8
- Layers = 4

---

# Physiological Interpretation

The Transformer can associate:

```
Frame 10
```

```
Frame 90
Frame 180
Frame 270
```

within the same heartbeat cycle.

This capability allows learning of physiological periodicity beyond local neighborhoods.

# 9. Feature Fusion Strategy

## Motion Features

Capture:

- Temporal changes
- Pulse-induced fluctuations
- Motion dynamics

---

## Appearance Features

Capture:

- Skin texture
- Anatomical structure
- Illumination cues

---

## Fusion Mechanism

$$
F_{fusion}
[F_{motion};F_{appearance}]
$$

Concatenated features are projected into a unified latent representation.

Benefits:

- Complementary information integration
- Improved robustness
- Better physiological discrimination

# 10. Spectral Attention Module

## Motivation

Heart-rate information primarily resides in specific frequency bands.

Standard Transformers operate in the time domain.

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## Spectral Attention Process

```
Temporal Features
       │
       ▼
FFT
       │
       ▼
Magnitude Spectrum
       │
       ▼
Attention Gate
       │
       ▼
Enhanced Features
```

Benefits:

- Frequency-aware learning
- Suppression of irrelevant frequencies
- Emphasis on cardiac harmonics

# 11. Temporal Refinement Block

After Transformer processing:

Local temporal inconsistencies may remain.

The system therefore applies:

- Depthwise Conv1D
- Pointwise Conv1D
- Residual connections

Benefits:

- Local smoothing
- Waveform continuity
- Improved pulse morphology

# 12. Loss Function Design

## Negative Pearson Loss

Measures waveform correlation:

[
Loss = 1-r
]

where:

[
r=
\frac{Cov(x,y)}
{\sigma_x\sigma_y}
]

Benefits:

- Shape preservation
- Scale invariance

---

## Smooth L1 Loss

Balances:

- L1 robustness
- L2 smoothness

Benefits:

- Stable training
- Reduced sensitivity to outliers

---

## Spectral Loss

Matches frequency-domain characteristics.

Benefits:

- Improved BPM estimation
- Better physiological frequency preservation

## SNR-Aware Loss

Encourages cleaner waveforms.

Benefits:

- Noise suppression
- Improved physiological quality

---

## Contrastive Temporal Loss

Encourages consistent physiological dynamics.

Benefits:

- Temporal discriminability
- Better sequence representation

---

## Why Multiple Losses?

Each loss targets a different aspect:

| Loss | Purpose |
|---|---|
| Pearson | Waveform shape |
| Smooth L1 | Amplitude accuracy |
| Spectral | Frequency fidelity |
| SNR | Signal quality |
| Contrastive | Temporal consistency |

Together they improve both waveform reconstruction and BPM estimation.

# 13. Training Strategy

## AdamW Optimizer

Advantages:

- Decoupled weight decay
- Better generalization
- Stable Transformer training

---

## Warmup Scheduler

Gradually increases learning rate.

Benefits:

- Prevents early instability
- Smooth optimization startup

---

## Cosine Annealing

Learning rate:

$$
\eta_t = \eta_{min} + \frac{1}{2} (\eta_{max}-\eta_{min}) (1+\cos(\pi t/T))
$$

Benefits:

- Improved convergence
- Better final minima

---

# Mixed Precision Training

Uses FP16 and FP32.

Benefits:

- Lower memory consumption
- Faster training

---

# Gradient Accumulation

Simulates larger batch sizes.

Benefits:

- Improved optimization stability
- Reduced GPU memory requirements

---

# Gradient Clipping

Limits exploding gradients.

Benefits:

- Stable Transformer training

---

# EMA Model Averaging

Maintains:

$$
\theta_{EMA}
\alpha\theta_{EMA}
+
(1-\alpha)\theta
$$

Benefits:

- Improved generalization
- Smoother convergence

---

# torch.compile Optimization

Benefits:

- Kernel fusion
- Reduced overhead
- Faster execution

# 14. Evaluation Metrics

## MAE BPM

[
MAE

\frac{1}{N}
\sum |y-\hat y|
]

Measures average BPM error.

---

## RMSE BPM

[
RMSE=
\sqrt{
\frac{1}{N}
\sum(y-\hat y)^2
}
]

Penalizes large errors.

---

## Pearson Correlation

Measures waveform similarity.

Range:

[
[-1,+1]
]

Higher is better.

---

## Signal-to-Noise Ratio

Measures physiological signal quality.

Higher SNR indicates:

- Cleaner waveform
- Better BPM reliability

# 15. Visualization and Experimental Analysis

## Total Loss Curve

Monitors overall optimization.

---

## Individual Loss Components

Tracks contribution of:

- Pearson loss
- Spectral loss
- SNR loss

---

## Validation MAE

Measures BPM accuracy.

---

## Validation RMSE

Evaluates large-error behavior.

---

## Pearson Correlation Curve

Measures waveform reconstruction quality.

---

## Validation SNR

Assesses physiological signal quality.

---

## Scatter Plot

Predicted BPM vs Ground Truth BPM.

Ideal trend:

[
y=x
]

---

## Bland–Altman Plot

Measures agreement between:

- Ground truth BPM
- Predicted BPM

Widely used in biomedical validation.

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## Time-Domain Comparison

Compares waveform morphology.

---

## Frequency-Domain Comparison

Compares dominant physiological frequencies.

---

# 16. Engineering Optimizations and Improvements

## $O(N^3) \rightarrow O(N)$ Detrending

Problem:

Traditional detrending is computationally expensive.

Solution:

Linear-time detrending implementation.

Gain:

Significant preprocessing acceleration.

---

## In-Memory Dataset Caching

Problem:

Repeated disk access.

Solution:

Cache processed samples.

Gain:

Reduced I/O bottleneck.

---

## Ground Truth Pre-Caching

Problem:

Repeated loading of BVP files.

Solution:

Precompute and store signals.

Gain:

Faster epochs.

---

# Cached Hann Windows

Problem:

Repeated FFT window generation.

Solution:

Reuse cached windows.

Gain:

Lower CPU overhead.

---

# Cached Frequency Masks

Problem:

Repeated frequency-band computation.

Solution:

Precompute masks.

Gain:

Faster BPM estimation.

---

# Corrected Pearson Loss

Problem:

Improper masking behavior.

Solution:

Robust masked Pearson implementation.

Gain:

Improved optimization stability.

---

# Parabolic FFT Interpolation

Problem:

Discrete FFT resolution.

Solution:

Sub-bin peak refinement.

Gain:

Improved BPM precision.

---

# Attention Pooling

Problem:

Average pooling loses important regions.

Solution:

Learned pooling weights.

Gain:

Enhanced physiological feature extraction.

---

# Depthwise Separable Convolutions

Problem:

High computational cost.

Solution:

Depthwise + pointwise decomposition.

Gain:

Reduced parameters and FLOPs.

---

# Gradient Checkpointing

Problem:

High memory usage.

Solution:

Recompute intermediate activations.

Gain:

Larger models under limited memory.

---

# 17. Research Contributions

The proposed architecture advances beyond traditional rPPG systems through:

- Dual-stream physiological representation
- Multi-scale spatial learning
- CBAM attention integration
- Transformer temporal modeling
- Spectral attention enhancement
- Temporal refinement block
- Multi-objective physiological optimization

Compared with traditional POS/CHROM methods, the system learns robust representations directly from data.

Compared with CNN-only models, it captures long-range temporal dependencies.

Compared with Transformer-only models, it preserves rich spatial information through dedicated CNN encoders.

The architecture therefore combines the strengths of both paradigms.

# 18. Limitations

Current limitations include:

- Dependence on facial visibility
- Sensitivity to extreme motion
- Illumination variation challenges
- Dataset diversity limitations
- Computational cost of Transformer inference

Further evaluation on larger datasets is necessary.

# 19. Future Work

Potential research directions include:

- Vision Transformers (ViT)
- Swin Transformers
- Cross-attention fusion
- Self-supervised physiological learning
- Multi-dataset pretraining
- Real-time deployment
- Edge AI optimization
- Mobile inference acceleration
- Federated physiological learning

# 20. Conclusion

This project presents a complete end-to-end Remote Photoplethysmography system built upon a Dual-Stream CNN–Transformer architecture.

The framework combines:

- Advanced ROI extraction
- Physiological signal processing
- Multi-scale CNN feature learning
- Transformer temporal modeling
- Spectral attention enhancement
- Temporal waveform refinement

to reconstruct blood volume pulse signals and estimate heart rate from facial videos.

The integration of motion and appearance streams enables robust physiological representation learning, while Transformer self-attention captures long-range temporal relationships that conventional CNN architectures cannot effectively model.

Additional engineering innovations including EMA stabilization, spectral enhancement, attention pooling, depthwise separable convolutions, cached signal-processing components, and optimized training strategies further improve efficiency and performance.

The resulting architecture provides a strong foundation for future research in contactless cardiovascular monitoring, remote health assessment, telemedicine, and next-generation physiological sensing systems.

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## I. Software and Framework References

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## J. References Specific to This Project's Contributions

The proposed work draws inspiration from DeepPhys, PhysNet, and PhysFormer while extending them through:

• Dual-stream CNN feature extraction

• Multi-scale spatial encoders

• CBAM-enhanced physiological attention

• SE-based channel attention

• Spectral attention-guided temporal modeling

• Transformer-based long-range physiological dependency learning

• Temporal convolution refinement

• ROI quality-aware signal acquisition

• Robust BPM estimation using physiological constraints and parabolic FFT interpolation

The final architecture represents a hybrid physiological deep-learning framework integrating classical signal processing with modern CNN–Transformer methodologies for remote cardiovascular monitoring.